Post-Quantum Cryptography

Risk Assessment — Executive Summary

ACME Bank

Organisation: ACME Bank
Assets Scanned: 2
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Executive Summary

Quantum Risk Score

42

out of 100

MEDIUM

Key Findings

- 4 quantum-vulnerable cryptographic components identified
- 1 component(s) at Harvest-Now Decrypt-Later risk
- 0 finding(s) require immediate action
- Mapped to PCI DSS 12.3.3, SEBI CSCRF, NIST FIPS 203/204, DST Task Force, RBI Q-SAFE, CERT-In and DPDP Act 2023 §8

Compliance Status

Framework	Status	Notes
PCI DSS 12.3.3	PASS	Cryptographic inventory required by PCI DSS 12.3.3. 4 components documented.
SEBI CSCRF	PARTIAL	Inventory satisfies the CSCRF control. 4 quantum-vulnerable components must be prioritised for PQC migration (binding on SEBI REs since Jun 2025).
NIST FIPS 203/204	PLANNED	Reference standard. Begin migration planning.
DST Task Force	PASS	Recommendatory roadmap. CII target: cryptographic-asset inventory by 2027, CBOM in govt procurement from FY2027-28. Cryptographic inventory completed within last 180 days.
RBI Q-SAFE	MONITOR	RBI Q-SAFE Committee (formed 25 May 2026) reviewing financial-sector cryptography via CBOM + crypto-agility. Framework expected ~Nov 2026.
CERT-In	MONITOR	No CERT-In-specific PQC advisory confirmed. Monitor for future guidance.
DPDP Act 2023 §8	MONITOR	DPDP Act s.8 safeguards + penalties (up to Rs.250 crore) become enforceable 2027-05-13 (DPDP Rules 2025, Phase III). Not yet live.

Top 5 Priority Actions

#	Action	Timeline	Effort
1	Replace certificate with ML-DSA-65 at next renewal (FIPS 204)	PLANNED	2-4 hours at next renewal
2	Add X25519Kyber768 hybrid group alongside X25519	PLANNED	1-2 days (CDN config change)
3	Replace certificate with ML-DSA-65 at next renewal (FIPS 204)	PLANNED	2-4 hours at next renewal

About Post-Quantum Cryptography

Quantum computers running Shor's algorithm will break RSA and elliptic-curve cryptography. While no cryptographically relevant quantum computer exists today, adversaries are collecting encrypted traffic now for future decryption (Harvest-Now Decrypt-Later). NIST finalised post-quantum standards in August 2024. India's DST Task Force published a migration roadmap in February 2026. Migration typically takes 42–54 months — organisations should begin now.

Methodology & Provenance

Tool: SeQr scanner — sslyze 6.0 for passive TLS analysis and an OpenSSL 3.5+ ML-KEM ClientHello for **active** post-quantum detection (passive scanning alone cannot confirm hybrid PQC support).

Method: passive measurement plus an active key-exchange probe. No authentication, no exploitation, no data exfiltration — equivalent to a single ordinary client connection, rate-limited.

Coverage: TLS (443), SSH key exchange + host keys (22), mail STARTTLS (SMTP / IMAP / POP3), and database TLS (PostgreSQL).

Scan confidence: high = direct connection; medium = cloud-provider IP; low = CDN-terminated (origin configuration is not externally observable).

Not assessed: origin servers behind CDNs, application-layer data handling, and internal / non-public services.

Assessment date: 09 September 2026 17:40 UTC.

HNDL Disclaimer: HNDL exposure scores are predictive cryptographic risk metrics based on data-type endurance, not indicators of active data interception or confirmed network breaches.